

The Geography of Child Penalties and Gender Norms:

A Pseudo-Event Study Approach

Replication Guide

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Last update: August 2025

This document is a comprehensive replication guide and technical appendix to the paper *“The Geography of Child Penalties and Gender Norms: A Pseudo-Event Study Approach.”* It provides all instructions and explanations required to replicate the results presented in the paper, including descriptions of data sources, cleaning procedures, the construction of a pseudo-event study dataset via matching, and the exact specifications used to generate each figure and table. Although all replication code is written in R, the procedures are described in a way that allows implementation in any programming language. Information on data availability and technical specifications of the computing environment is provided in the README file.

The document is organized into three main sections, each with multiple subsections covering specific data or analytical steps:

1. **Data Cleaning:** Preparation of Current Population Survey (CPS), American Community Survey (ACS), Panel Study of Income Dynamics (PSID), National Longitudinal Survey of Youth (NLSY), and others.
2. **Pseudo-Event Study Methodology:** Implementation of the matching algorithm and construction of pseudo-panels, applied to CPS and ACS, and validated with PSID and NLSY data.
3. **Results:** Instructions and specifications to reproduce all figures and tables, linking cleaned data and matched samples to the full set of analyses.

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1. Data Cleaning

1.1 Current Population Survey (CPS)

The raw data file is called "raw_cps.RData." CPS variable coding and details can be found at [IPUMS CPS](#).

1.1.1 Sample Selection

We combine Basic Monthly Files from 1989-2020 and the Annual Social and Economic Supplement (ASEC) from 1968-2020.

The sample used in the matching procedure is composed of

1. Childless people aged 20-45 in the years 1968-2020
2. Parents with age at first birth 25-45 in the years 1973-2020 interviewed between event times 0-10 (event times will be defined later in the cleaning description)

Additionally, observations with a zero, negative or missing weight, missing state FIPS code, unknown parental status, missing educational attainment, or missing marital status are dropped. According to IPUMS, all Basic Monthly observations in the month of March are included in the ASEC sample, along with an ASEC oversample, so we also drop Basic Monthly observations in March (as indicated by ASECFLAG) to avoid double counting.

1.1.2 Variables Created

The following variables are created after cleaning the CPS

1. Demographic variables
 - (a) Age
 - (b) Year of birth
 - (c) Indicator for female
 - (d) Education level: Below high school, high school, some college, college and above

- (e) Indicator for college and above
- (f) Indicator for high school or below
- (g) Marital status: Married, spouse present; Married, spouse absent/Separated; Divorced; Widowed; Never married/single
- (h) Indicator for married (1 if Married, spouse present or Married, spouse absent, 0 otherwise)
- (i) Race/ethnicity: White, non-hispanic; Black, non-hispanic; Other; Hispanic
- (j) Race: White, Black, Other (includes Hispanic respondents in each of the categories rather than as a separate category)
- (k) Indicator for white respondents: 1 if race == "White", 0 otherwise
- (l) Indicator for white, non-hispanic respondents: 1 if race/ethnicity == "White, non-hispanic", 0 otherwise
- (m) Indicator for Black respondents: 1 if race == "Black", 0 otherwise
- (n) Indicator for Black, non-hispanic respondents: 1 if race/ethnicity == "Black, non-hispanic", 0 otherwise

2. Children

- (a) Number of children
- (b) Indicator for any children
- (c) Age of oldest child in household
- (d) Age at first birth: current age - age of oldest child in household
- (e) Year of birth of first child: current year - age of oldest child in household
- (f) Decade of birth of first child: groups the year variable into decades (1973-79, 1980-89, 1990-99, 2000-09, 2010-20), can exclude births in previous decades or create separate levels for them, but the above levels should be defined as stated (eg: do not create a level for all of 1970-79) and we will not be using the previous decades.

3. Geographic variables

- (a) State of residence: FIPS codes and names for post-1977 data, state groups for 1968-1976
- (b) State groups: Use the state groups provided in the CPS state FIPS variable before 1976 and create a variable with these groupings for the whole period. The state groups FIPS codes are also not consistent between 1973 and 1976 so we need to combine them into the most granular state group

that can be observed in the whole period. For example, between 1973 and 1976 we have both “Alaska-Washington-Hawaii” and “Washington-Oregon-Alaska-Hawaii” available so our state group variable will be defined as “Washington-Oregon-Alaska-Hawaii” for 1972-2019.

- (c) Census division groups (residence): constructed based on the 9 categories listed here (New England, Middle Atlantic, East North Central, West North Central, South Atlantic, East South Central, West South Central, Mountain, Pacific) [official Census Bureau definition](#)
- (d) Election results in state of residence: group observations into red states (if red in last 4 elections), blue states (if blue in last 4 elections), swing states (neither red nor blue in all 4 elections) [[we use the following map](#)]
- (e) Metro status: metro if METRO takes on value 2, 3, or 4 and non-metro if METRO takes on value 1, missing otherwise
- (f) Country of birth: clean BPL variable which contains country of birth (using names is a bit easier here since the numeric codes using in CPS and ACS are not the same for foreign countries). You should clean all the countries (although it can be quite messy), but the countries we are interested in are: Argentina, Afghanistan, Algeria, Australia, Bangladesh, Belgium, Bolivia, Brazil, Bulgaria, Canada, Chile, China, Colombia, Costa Rica, Cuba, Czech/Slovakia (group Czech Republic, Slovakia and Czechoslovakia), Dominican Republic, Ecuador, Egypt, El Salvador, Ethiopia, France, Germany, Greece, Guatemala, Guyana, Haiti, Honduras, Hong Kong, Hungary, India, Indonesia, Iran, Iraq, Ireland, Israel (group Israel and Israel/Palestine), Italy, Jamaica, Japan, Jordan, Kenya, South Korea (if there is a broad “Korea” category, assume this is South Korea), Laos, Mexico, Morocco, Netherlands, Nicaragua, Nigeria, Peru, Philippines, Poland, Portugal, Puerto Rico, Romania, Russia (there is no separate Russia category so assume any category that isn’t another country under the USSR umbrella such as “Other USSR/Russia”, “USSR, ns” are Russia), South Africa, Spain, Sweden, Switzerland, Taiwan, Thailand, Trinidad and Tobago, Turkey, UK, Venezuela, Vietnam). Some other cleaning decisions for countries that we aren’t currently using: “Congo”, “Zaire” == Democratic Republic of Congo. We also add the United States as a category to identify US-born people, although in the CPS we cannot identify their state of birth which we can in the ACS.

4. Work and income

- (a) Earnings in previous year: INCWAGE
- (b) Annual employment in previous year: takes on a value of 1 if INCWAGE is non-zero, takes on a value of 0 if INCWAGE is zero

- (c) Weekly employment: takes on a value of 1 if the respondent is in the armed forces (EMPSTAT == 1), was at work (EMPSTAT == 10), was not at work last week but has a job (EMPSTAT == 12) and takes on a value of 0 if the respondent is unemployed (EMPSTAT == 20-22) or not in the labor force (EMPSTAT == 30-36)

5. Interview details

- (a) ID for individual: Identifier construction when combining Basic Monthly data and March files is relatively complicated, but we use YEAR, SERIAL, PERNUM to uniquely identify individuals who are in Basic Monthly files or in March files but not in the ASEC oversample (indicated by ASECTAG and ASECOVERP) and use a combination of YEAR, SERIAL, PERNUM and CPSIDP for those in the oversample. We want to track individuals across months and link them between the Basic Monthly and March files as accurately as possible.
- (b) Weights: Use ASECWT for ASEC observations and WTFINL for Basic Monthly observations.
- (c) Interview year
- (d) Five-year interval: Groups the year variable into intervals of five years (1973-79, 1980-84, 1985-89, 1990-94, 1995-99, 2000-04, 2005-09, 2010-20)
- (e) Decades: Groups the year variable into decades (1973-79, 1980-89, 1990-99, 2000-09, 2010-20)
- (f) Indicator for CPS data

1.2 American Community Survey (ACS)

The raw data file is called "raw_acs.RData". Variable coding and details can be found at [IPUMS USA](#).

1.2.1 Sample Selection

The sample used in the matching procedure is composed of

1. Childless people aged 20-45
2. Parents with age at first birth 25-45 interviewed between event times 0-10 (event times will be defined later in the cleaning section)
3. In the years 2000-2019

Additionally, observations with a zero, negative or missing weight, missing state FIPS code, unknown parental status, missing educational attainment, or missing marital status are dropped.

1.2.2 Variables Created

The following variables are created after cleaning the ACS. Note that cleaning the ACS is very similar to cleaning the CPS but there are some minor differences to keep track of.

1. Demographic variables

- (a) Age
- (b) Year of birth
- (c) Indicator for female
- (d) Education level: Below high school, high school, some college, college and above
- (e) Indicator for college and above
- (f) Indicator for high school or below
- (g) Marital status: Married, spouse present; Married, spouse absent/Separated; Divorced; Widowed; Never married/single
- (h) Indicator for married (1 if Married, spouse present or Married, spouse absent, 0 otherwise)
- (i) Race/ethnicity: White, non-hispanic; Black, non-hispanic; Other; Hispanic
- (j) Race: White, Black, Other (includes Hispanic respondents in each of the categories rather than as a separate category)
- (k) Indicator for white respondents: 1 if race == "White", 0 otherwise
- (l) Indicator for white, non-hispanic respondents: 1 if race/ethnicity == "White, non-hispanic", 0 otherwise
- (m) Indicator for Black respondents: 1 if race == "Black", 0 otherwise
- (n) Indicator for Black, non-hispanic respondents: 1 if race/ethnicity == "Black, non-hispanic", 0 otherwise

2. Children

- (a) Number of children
- (b) Indicator for any children
- (c) Age of oldest child in household
- (d) Age at first birth: current age - age of oldest child in household

- (e) Year of birth of first child: current year - age of oldest child in household
- (f) Decade of birth of first child: groups the year variable into decades (1973-79, 1980-89, 1990-99, 2000-09, 2010-20), can exclude births in previous decades or create separate levels for them, but the above levels should be defined as stated (eg: do not create a level for all of 1970-79) and we will not be using the previous decades.

3. Geographic variables

- (a) State of residence: FIPS codes and names
- (b) Census division of residence: constructed based on the [official Census Bureau definition](#), same as for CPS
- (c) Election results: group observations into red states (if red in last 4 elections), blue states (if blue in last 4 elections), swing states (neither red nor blue in all 4 elections) [[we use the following map](#)]
- (d) Metro status: metro if METRO takes on value 2, 3, or 4 and non-metro if METRO takes on value 1, missing otherwise
- (e) Country of birth: clean BPLD variable which contains state of birth if born in US (unlike CPS) or country of birth if born abroad (using names is a bit easier here since the numeric codes using in CPS and ACS are not the same for foreign countries). You should clean all the countries (although it can be quite messy), but the countries we are interested in are: Argentina, Afghanistan, Algeria, Australia, Bangladesh, Belgium, Bolivia, Brazil, Bulgaria, Canada, Chile, China, Colombia, Costa Rica, Cuba, Czech/Slovakia (group Czech Republic, Slovakia and Czechoslovakia), Dominican Republic, Ecuador, Egypt, El Salvador, Ethiopia, France, Germany, Greece, Guatemala, Guyana, Haiti, Honduras, Hong Kong, Hungary, India, Indonesia, Iran, Iraq, Ireland, Israel (group Israel and Israel/Palestine), Italy, Jamaica, Japan, Jordan, Kenya, South Korea (if there is a broad "Korea" category, assume this is South Korea), Laos, Mexico, Morocco, Netherlands, Nicaragua, Nigeria, Peru, Philippines, Poland, Portugal, Puerto Rico, Romania, Russia (there is no separate Russia category so assume any category that isn't another country under the USSR umbrella such as "Other USSR/Russia", "USSR, ns" are Russia), South Africa, Spain, Sweden, Switzerland, Taiwan, Thailand, Trinidad and Tobago, Turkey, UK, Venezuela, Vietnam). Some other cleaning decisions for countries that we aren't currently using: "Congo", "Zaire" == Democratic Republic of Congo. We also add the United States as a category to identify US-born people, although in the CPS we cannot identify their state of birth which we can in the ACS.
- (f) State of birth: create a state of birth variable using the BPLD variable, identifying the specific state that people born in the United States were born in.

- (g) Census division of state of birth: use BPLD and group states into census divisions just as for states of residence
 - (h) Country of ancestry: We are interested in getting a variable with the same 81 countries that we have the country of birth variable for. Clean ANCESRTR1 which really reports ethnicity of ancestry rather than country so that we have matching categories. This can be quite messy to clean and there are some decisions that are not clear cut about which group certain categories should be put in so we may have to discuss at some point.
4. Work and income
- (a) Earnings in previous year: INCWAGE
 - (b) Annual employment in previous year: takes on a value of 1 if INCWAGE is non-zero, takes on a value of 0 if INCWAGE is zero
 - (c) Weekly employment: takes on a value of 1 if the respondent is employed ($EMPSTAT == 1$) or a value of 0 if the respondent is unemployed ($EMPSTAT == 2$) or not in the labor force ($EMPSTAT == 3$)
5. Interview details
- (a) ID for individual: observations are uniquely identified by YEAR, SERIAL, and PERNUM
 - (b) Weights: Use PERWT
 - (c) Interview year
 - (d) Five-year interval: Groups the year variable created above into intervals of five years (1973-79, 1980-84, 1985-89, 1990-94, 1995-99, 2000-04, 2005-09, 2010-20)
 - (e) Decades: Groups the year variable into decades (1973-79, 1980-89, 1990-99, 2000-09, 2010-20)
 - (f) Indicator for ACS data

1.3 Panel Study of Income Dynamics (PSID)

We use a combination of variables from the Family-level and Individual-level PSID data and we also use the Childbirth and Adoption History data provided in the Family History files. The Childbirth and Adoption History file can be found in "Raw_Data\PSID\cah85_17" and needs to be merged with the Family and Individual-level file. The individual/family-level file with the main data we need is called "raw_psid.RData" and variable coding and details can be found on [the University of Michigan's Institute of Social Research PSID website](#)

1.3.1 Sample Selection

The sample used is composed of

1. Household heads and their spouses
2. who have available earnings or employment data,
3. have non-missing and non-zero sample weights,
4. are in the core sample, Latino sample, or immigrant sample
5. are interviewed in 1968-1997 and two-year intervals from 1999-2017
6. drop those who have a missing date of birth (both month of birth and year are missing)
7. drop those with missing education, marital status, race, state (once they have been cleaned)

For the PSID and NLSY pseudo-panels/matching procedure we add the following (filename `nlsy79_clean_all.rds`):

1. have an annual or weekly event time between 0 and 10 and an age at first birth between 25 and 45 for parents, a missing event time, any event time less than 0 (Note that this means that no matter how negative their event time is, we keep them in the sample)

For the PSID and NLSY actual event studies we add the following (filename `nlsy79_clean_restricted.rds`):

1. have an annual or weekly event time between -5 and 10 (event times are defined below)
2. with an age at first birth between 25 and 45

1.3.2 Variables Created

Unless a single year is specified, all variables are included for all years they are available for since PSID provides each variable separately by year in a wide format rather than a long format and variables need to be renamed (names are usually not informative of either the content or the year) and matched across years when cleaning. Many of the variables we use are reported at the household level but are reported for both the household head and the spouse, so we can use these to produce individual-level observations for the head and spouse, but not for other household members. Often, individual and household-level options are available and we may need to select the best option between them. Since this is a panel dataset, we also want to correct as much as possible

for consistency over time for variables that should be time invariant. For example, date of birth should not vary for any individual and we generally choose the date of birth among each individual that is most commonly reported and then correct for incorrect age values using the chosen date of birth. We want to correct variables like race and state as well. Note that whenever the variable is constructed by taking the mode we prioritize the minimum in the case of ties.

1. Childbirth and Adoption History file: merge with the rest of PSID data after cleaning to add date of birth of oldest child to the main PSID dataset.

(a) Sample selection: Keep biological children only (CAH2 (Record Type) == 1)

(b) Variables:

- i. Month of birth of child: use CHILD_DOBM, replace with approximate month if exact month is unknown. If winter is reported (CHILD_DOBM == 21), assume January; if spring is reported (CHILD_DOBM == 22), assume April; if summer is reported (CHILD_DOBM == 23), assume July; if season is also unknown, assume September.
- ii. Year of birth of child: use CHILD_DOBY
- iii. Date of birth of child: create using CHILD_DOBM and CHILD_DOBY
- iv. Birth order: For each parent, generate a new birth order by the cleaned date of birth variable (with missings replaced) rather than the variable provided in the data after adopted children have been used and then use this new birth order variable to determine which child is the oldest and what their date of birth is.
- v. Total number of children: As with birth order, create a new variable that counts the total number of biological children for each parent.

2. Interview details

(a) Individual ID number in 1968: ER30002

(b) Household ID number in 1968: ER30001

(c) Unique ID number: Combine ER30001 and ER30002 to create a unique identifier

(d) Sequence number

(e) Relationship to household head

(f) Weights: there are various options for which weights to use in order to create a single weight variable that is consistent across years. Presently, we use Individual weights wherever possible (1968-1989) and for other years use individual combined latino and core weights (1990-1996) and combined longitudinal weights (1997-2017)

- (g) Month of interview: from 1968-1979 dates are in ranges of 15 days rather than exact (eg: V99 == 7 indicates an interview somewhere between May 27 and June 19) so we create an interview month variable that uses the month in which most days in the range are (eg: month of interview == June when V99 == 7). Note that the coding is not consistent across all years up to 1979 either. From 1989 to 1996 the exact day and month are reported together in the same variable so we can extract the month from this. From 1997 onwards we have a different interview month variable which reports the month of interview on its own. If month of interview is missing, we replace it with the most commonly reported month in the data which is April.
- (h) Date of interview: combine month and year of interview, assume the day of interview is always the 15th
- (i) Indicator for household head: if they are reported as the household head in a given year and their sequence number is one
- (j) Indicator for spouse: if they are reported as the husband or wife of the household head and their sequence number is two
- (k) Indicator for core sample observation: their household ID number is between 1 and 2930
- (l) Indicator for immigrant sample: household ID number between 3001 and 4851
- (m) Indicator for census sample: household ID number between 5001 and 6872
- (n) Indicator for latino sample: household ID number between 7001 and 9308
- (o) Household size
- (p) Indicator for PSID data

3. Demographic variables

- (a) Age: Replace PSID's age variable with imputed age instead since there are many missings for individuals and some values that appear to be incorrect (take the number of months between date of interview and date of birth divided by 12, floored rather than rounded up).
- (b) Year of birth: take the most commonly reported year of birth across interview years for each individual
- (c) Month of birth: take the most commonly reported month of birth across interview years for each individual. If someone is missing their month of birth but not the year after cleaning, assume they were born in September
- (d) Date of birth: combine year and month of birth and assume day is the 15th for everyone

- (e) Age at first birth: create it using the date of birth of the parent and the date of birth of their first child (months between date of parent's birth and date of first childbirth divided by 12 and floored rather than rounded). Treat as missing if age at first birth is less than 14.
- (f) Indicator for having a child: takes on a value of 1 if the date of birth of first child variable from the Childbirth and Adoption History file is non-missing and 0 if it is missing.
- (g) Indicator for female
- (h) Education level: Below high school, high school, some college, college and above
 - i. Use individual-level years of education variable and take the maximum over all years it is non-missing. Then use the family-level education variable and create a new variable which takes the head variable if the person is identified as a household head or the spouse variable if they are a spouse . Get the maximum years of education using this variable as well. Highest year of education == max based on the individual-level variable as long as it is non-missing. If it is missing, fill it in with the family-level max. Education level == "Below HS" if the maximum reported years are under 12, "HS grad" if they report exactly 12 years, "Some college" if they report between 13 and 15 years, "College and above" if they report between 16 and 17.
- (i) Marital status: Married, spouse present; Married, spouse absent/Separated; Divorced, Widowed, Never married/single
 - i. Use individual-level variable rather than head vs spouse variables
- (j) Indicator for married (1 if Married, spouse present or Married, spouse absent, 0 otherwise)
- (k) Race: White, non-hispanic; Black, non-hispanic; Other; Hispanic
 - i. Create a single race variable for race of the individual based on the head and spouse race variables that takes on the value of the head variable if the person is a head in a given year or the value of the spouse variable if they are a spouse in that year after having cleaned the head and spouse variables so they are consistent. For each individual take the most commonly reported race across all survey years. Create a separate Latino indicator using the Latino codes that are included in the head and spouse race variables for some years and the head and spouse hispanicity variables (someone is Latino in a given year if any of these 4 variables defines them as such). If the individual is defined as Latino in any year, we treat them as Hispanic regardless of the mode of the race variable on its own. Note that the coding for race variables is quite inconsistent over time so this needs to be corrected for before taking the mode.

Note: many of these variables are inconsistent over time in terms of coding and reporting error, so we need to do a fair amount of cleaning. For variables like age, we also need to correct them

if there are obvious errors like age decreasing over time. For variables like race that are time invariant we also have different races reported for the same individual depending on the year, so we also take the most commonly reported value for each value.

4. Work and income

(a) Earnings in previous year:

(b) Note: Missing value codes change a lot from year to year and across all the various income variables. General rules are that codes for “no income” from any activity should be treated as 0 rather than as missing, any codes that specifically indicate income is not known or not recorded for someone should be treated as missing. I have specified what I recode based on the codebook but it’s very easy to make a mistake on this so you should check the codebook yourself and let me know if you disagree with any of the codes or years I have identified.

i. Head: Use head’s earnings in previous year variable until 1993, in later years add head’s earnings variable to head’s business earnings and half of farm income if marital status is married, spouse present. If unmarried/married, spouse absent, in later years add head’s earnings, head’s business earnings and all of household farm income. If any of the three earnings values that are being added together are missing, we treat them as a zero. If all three are missing, we treat earnings as missing.

A. *Earnings in previous year including business and farm*: nothing to recode [Note that this is often topcoded which is fine, we don’t want the topcode to be missing]; *Earnings in previous year excluding business and farm*: Recode 9,999,999 in 1994 and 1995 (Latino sample family code); *Business income*: Recode 999,999 in 1994, 1995, 1997 (Latino or Immigrant sample family code) [Note that I only clean this variable for 1993 onwards so I have not specified anything that might be necessary for earlier years].

ii. Spouse: Use wife’s earnings in previous year variable until 1993, add wife’s earnings excluding business and farm, wife’s business income and half of household’s farm income from 1994 onwards to get wife’s total earnings from 1993 onwards. If any of the three earnings values that are being added together are missing, we treat them as a zero. If all three are missing, we treat earnings as missing.

A. *Earnings in previous year including business and farm*: nothing to recode; *Earnings in previous year excluding business and farm*: in 1993 and 1994 recode 9,999,999 (Latino sample family) [Note that there are many years where 0 may indicate there was no wife in the family unit which is fine to leave as a 0 because we will not be using these spouse income variables

if there is no spouse anyway]; *Business earnings*: in 1994, 1995, and 1997 recode 999,999 (Latino sample sample) [Again, note that 0 may indicate that there was no wife in the family unit or they did not own a business which is fine to leave as a 0 rather than recode to missing. In some years 0 may indicate the business lost money, and I think a 0 is also fine here since we can't distinguish between whether it is 0 because they actually lost money or if it's a true zero].

- iii. Missing values for household farm income: Recode -999,999 in 1993 (Loss of an unknown amount), 999,999 in 1994 and 1995 (Latino sample family) [Note that in many years there is a bottom coded value for a loss of a certain amount or more and we keep this, since we know it is a loss of at least that much and want to take this account when we sum the income variables even though we might be understating the loss]
 - iv. Create a single variable that takes on either the value of head's earnings as created above or spouse's, depending on whether the individual is a household head or spouse.
- (c) Annual employment in previous year: takes on a value of 1 if earnings are non-zero, takes on a value of 0 if earnings are zero, missing if earnings are missing
- (d) Weekly employment: If the individual is a household head and the head employment status variable is non-missing we use it and do the same for the spouse and for any leftover missing values we use the individual-level variable. Generate a dummy variable that takes on a value of 1 if the employment variable indicates they are "Working" or "Temporarily laid off" and is 0 otherwise (as long as the employment variable is non-missing). We treat employment as missing for anyone under 16 or older than 99.

5. Geographic variables

- (a) State: Combine GSA codes and FIPS codes. If FIPS is available, we use that. If it is missing, we use GSA instead. If both are missing, we replace the individual's state value in that year with their most commonly reported FIPS code across all years. If they never have a reported FIPS code, we then use their most commonly reported GSA code across all years.

6. Event times: We need two different variables since Weekly Employment refers to the time of interview and Earnings/Annual Employment refer to the previous year

- (a) Weekly Employment: We define event time $t = 0$ to start 6 months before the birth of the first child and to end 5 months after. $t = -1$ begins 12 months before the start of $t = 0$ and so on until $t = -5$, $t = 1$ begins 12 months after the end of $t = 0$ and so on until $t = 10$. For example, if a child is

born in July 1996, $t = 0$ begins in February 1996 and ends in January 1997. Weekly Employment observations from any parents with a child born in July 1996 who are interviewed in this window are assigned to $t = 0$.

- (b) Earnings: Rather than simply taking the event times we define for Weekly Employment and assigning earnings from the calendar year before, we want to take these event times and assign earnings from the calendar year in which most of the event time was realized. Using the example from above, most of $t = 0$ takes place in the year 1996, so we would like to assign earnings in calendar year 1996 to $t = 0$. Since Earnings/Annual Employment observations reported in 1997 will refer to 1996, they are assigned to $t = 0$ for all parents with a child born in July 1996. We continue on in the same way for all other event times.

1.4 National Longitudinal Survey of Youth - 1979 Cohort (NLSY)

The raw data file is called "raw_nlsy.RData". Variable coding and details can be found at [NLS Investigator](#), where you want to select NLSY79 and upload the tagset "NLSY79.NLSY79" in the Raw Data folder to find variable coding and details.

1.4.1 Sample Selection

We keep observations with

1. Non-missing interview dates
2. Non-missing and non-zero sample weights
3. Non-missing weekly employment or earnings values
4. Drop those with missing education, marital status, race, state (once they have been cleaned)

For the PSID and NLSY pseudo-panels/matching procedure we add the following (filename nlsy79_clean_all.rds):

1. have an annual or weekly event time between 0 and 10 and an age at first birth between 25 and 45, a missing event time, or any event time less than 0 (Note that this means that no matter how negative their event time is, we keep them in the sample)

For the PSID and NLSY actual event studies we add the following (filename nlsy79_clean_restricted.rds):

1. have an annual or weekly event time between -5 and 10 (event times are defined below)
2. with an age at first birth between 25 and 45

1.4.2 Variables Created

Unless a single year is specified, all variables are included for all years they are available for since NLSY provides each variable separately by year in a wide format rather than a long format and variables need to be renamed (names are usually not informative of either the content or the year) and matched across years when cleaning.

1. Interview details

- (a) Individual ID number in 1979
- (b) Interview date: Combine interview year and month variable, if month is missing assume it is March
- (c) Weights: Use the variable SAMPWEIGHT divided by 100 since it has 2 implied decimal places
- (d) Indicator for NLSY data

2. Childbirth

- (a) Date of birth of oldest child: Assume September if the month of birth is unknown but the year is non-missing
- (b) Number of children

3. Demographic variables

- (a) Age
- (b) Year of birth
- (c) Month of birth: Assume March if unknown
- (d) Date of birth: Combine year and month of birth, assume date is 15th
- (e) Indicator for female
- (f) Race: White, non-hispanic; Black, non-hispanic; Hispanic; Other, created using a combination of SAMPLE_RACE and FAM-30_1 (variable for 1st or only racial origin) which contains information on country of ancestry rather than race which we use to infer racial group (eg: european descent is white, non-hispanic, there is a specific code for black respondents etc.)
- (g) Marital status: Married, living together; Married, living apart; Divorced; Widowed; Never married/single using MARSTAT_KEY rather than the collapsed version.

- (h) Education level: Combine questions that ask “What is the highest degree you have received since last interview date?”, “What is the highest degree you have ever received” and “What is the highest grade you have completed” (Q3-10B, Q3-10D, and HGCREV) to create a single variable with the following categories: Below high school (highest grade < 12 and highest degree == none or unknown), HS degree (highest degree == highest degree == high school diploma or equivalent or if highest degree is none/unknown if highest grade == 12), Some college (highest degree == associate’s degree/junior college or if highest degree is none/unknown if highest grade is 13-15), college and above (highest degree == bachelor’s master’s degree, doctoral degree, or professional degree or if highest degree is none/unknown if highest grade is higher than 15). To deal with changing responses over time we base this off the highest degree or grade reported for an individual across all years that they are interviewed. Make sure to recode highest degree variable == 8 (other) as missing before taking the max across all years. Note that Q3-10B and Q3-10D have some years where both options are available, and in those years we simply include both variables and take the max over all the Q3-10B and Q3-10D years to get the highest degree received by each person.

4. Work and income

- (a) Weekly Employment: we use the variable that reports labor force status (prefix “STATUS_WK”) in each week for every week of the year. We create one variable that extracts the labor force status for the week of interview in a given year to the best of our ability. We are only provided the month and year of interview and not the exact date, so we assume interviews occur in the middle of the month. We then create a dummy variable that takes on a value of 1 if the labor force status variable in the week of interview reports a survey round or job number (greater than or equal to 100) or if they are in active military service (==7) and 0 if they are not working (==2), associated with employment but all time is not accounted for (==3), unemployed (==4), out of the labor force (==5). Weekly Employment is treated as missing if no information is reported for that week (==0).
- (b) Earnings: We use variables that report total income from wages and salary in the past calendar year, although there are slight variations in naming and measurement across years. From 1979-81 we use the standard version, while for 1982-2000 the closest available variable truncates earnings for the top 2% and revises some errors. Similarly, we use variables that are truncated for the top 2% for 2002-2014, but that are not revised.
- (c) Annual Employment: takes on a value of 1 if earnings are non-zero, takes on a value of 0 if earnings are zero

5. Event times: We need two different variables since Weekly Employment refers to the time of interview

and Earnings/Annual Employment refer to the previous year

- (a) Weekly Employment: We define event time $t = 0$ to start 6 months before the birth of the first child and to end 5 months after. $t = -1$ begins 12 months before the start of $t = 0$ and so on until $t = -5$, $t = 1$ begins 12 months after the end of $t = 0$ and so on until $t = 10$. For example, if a child is born in July 1996, $t = 0$ begins in February 1996 and ends in January 1997. Weekly Employment observations from any parents with a child born in July 1996 who are interviewed in this window are assigned to $t = 0$.
- (b) Earnings: Rather than simply taking the event times we define for Weekly Employment and assigning earnings from the calendar year before, we want to take these event times and assign earnings from the calendar year in which most of the event time was realized. Using the example from above, most of $t = 0$ takes place in the year 1996, so we would like to assign earnings in calendar year 1996 to $t = 0$. Since Earnings/Annual Employment observations reported in 1997 will refer to 1996, they are assigned to $t = 0$ for all parents with a child born in July 1996. We continue on in the same way for all other event times.

1.5 Census Bureau Population Data (2019)

For weights for some specification, we require population weights by state using current US population data. The file "nst-est2019-alldata.csv" is used to calculate the fraction of people in the US in 2019 that live in each US state. The final dataset should contain the name of the state and/or the fips code and the fraction.

1.6 General Social Survey Data (GSS)

There are two raw datasets which contain the GSS data. *GSS7218* contains the raw GSS data with all the variables, but does not contain state identifiers. *cross7316geo* contains the state identifiers. In order to get a complete raw dataset of the GSS merge these two using the variables *id* and *year*.

1.6.1 Sample Selection

We use the GSS data from 1973 to 2018 but exclude the observations oversampled in 1982 and 1987 survey rounds.

1.6.2 Variables Created

1. General variables

- (a) Year (*year*)
- (b) ID (*id*)
- (c) State (*fipsstat*)
- (d) Variable Weights (*wtssall*)

2. Norms variables

- (a) It is much better for everyone involved if the man is the achiever outside the home and the woman takes care of the home and family (*fefam*)
- (b) A working mother can establish just as warm and secure a relationship with her children as a mother who does not work (*fechld*)
- (c) A preschool child is likely to suffer if his or her mother works (*fepresch*)

1.6.3 Creating the Gender Progressivity Index (GPI)

To create the gender progressivity index from the three norms variables, follow these steps:

1. Rescale each norms variable so that a higher value is more gender progressive.
2. Standardize each norms variable to have mean 0 and standard deviation 1 across the entire sample.
3. Calculate the mean of these three standardized variables for each individual to get the GPI.

We also calculate the average of gender progressivity index by state, by time, and by state and decade. This is just a weighted average using the weights variable *wtssall*. The file named *gss_clean.rds* contains this standardized gender progressivity index (called *index_ad*)

2. Pseudo-Event Study Methodology

The pseudo-event study is constructed by assigning negative event times through matching childless individuals to recent parents, while positive event times come directly from observed parents. Matching is performed separately for Weekly and Annual samples within bins of demographic and geographic variables, with tie weights ensuring balanced pseudo-panels. The procedure accounts for survey-specific features (e.g., CPS pre-1977 state groups, ACS early years) and is validated against panel data from PSID and NLSY.

2.1 Matching (CPS and ACS)

Description of Procedure

- We use the `Match` function in the R package [Matching](#).
 - We use `Matching::Match` *only* to assign matching weights and ids for each matched pair. The matching design itself is defined by the bins created with `plyr::dply`.
 - The function `MatchBy` in the same package sounds ideal for our purposes but is incredibly slow; therefore we split the data into the bins we want to match within and iterate `Match` over those bins, which is significantly faster than running `Match` on the full dataset.
- Using cross-sectional data we are unable to observe people who do in fact have children in the years before they have their child. This is problematic for performing an event-study analysis, but we are able to produce a pseudo-event study by assigning negative event times to childless people by matching them to recent parents (parents with a child aged 0). For positive event times, we can simply use the observations we have of true parents.
- The matching is conducted separately for ACS and CPS data and we also split annual and weekly samples for each dataset and run a separate matching (4 different matchings total). This is because in CPS earnings and annual employment are only available in March/in the ASEC supplement and in ACS we have some observations with only one of earnings/annual employment or weekly employment available. To create a balanced pseudo-panel and maximize the observations available for each variable, we want to only match parents with available annual variables to childless people with available annual variables even if weekly

employment is missing and do the same for weekly employment. For each matching, drop missing values of the dependent variable of interest before running it.

- A given parent with a first child aged 0 is matched to one childless person of the same gender, education level, race (the race/ethnicity variable with Hispanic as a separate category), marital status, living in the same state who is one year younger and interviewed one year before. This gives us an observation at event time $t = -1$. We continue the same process, matching on the same demographics to someone who is two years younger and interviewed two years before to get an observation at $t = -2$ and so on for $t = -3, t = -4$, and $t = -5$. We use parents with children aged 1–10 in all the event-study regressions (event times 1–10) but we do not match them to any childless people.
 - Rather than actually matching to one person only, we match with ties so that whenever there are multiple childless matches available, the parent is matched to all the available matches, but they are weighted according to the number of childless matches. So a given parent may be matched to n childless people to get event time t and each childless person will be treated as $1/n$ observations rather than 1 whole observation. The reason for matching with ties is to eliminate the effect of the seed and the random element of selecting only one childless match when there are many available.
- Any childless people not matched to at least one parent are dropped, but any parents who we are unable to find a match for are kept in the data, although you will need to keep track of which parents are unmatched at each event time to create one of the descriptive statistics tables later.
- Sample: as described in Data cleaning section
 - Parents: age at first birth between 25 and 45, interviewed 5 or more years after the first available interview year in the data in order to create a balanced panel. For example, since the CPS data begins in 1968, we only keep parents interviewed 1973 onwards so that we do not have parents in 1969 with matches available only for $t = -1$ in 1968 and parents in 1970 with matches available only for $t = -1$ and $t = -2$ and so on.
 - Childless individuals: by restricting age and interview year for parents and matching on them, we automatically keep childless individuals aged 20 to 45 in all available interview years.
- **A few important caveats:**
 - Since Earnings and Annual Employment (which we use the CPS ASEC/March files matching for) are recorded for the year prior to any given interview year, a parent observed with a child aged 0 is in fact reporting their earnings at event time $t = -1$. We choose not to use these observations in our event study and exclusively use childless matches in the pre-period. Therefore, when we match within this

sample, we match parents with a child aged 0 to a childless person of the same age interviewed in the same year to get an observation at event time $t = -1$ and continue the same way until we match to a childless person 4 years younger interviewed 4 years ago to get an observation at event time $t = -5$. The people used in the matching with a child aged 0 will actually not appear anywhere in our event study.

- In the CPS data, state FIPS codes for every state are not reported until 1977, although we can observe the region an individual lives in (e.g.: Washington–Oregon–Alaska–Hawaii and Michigan–Wisconsin are listed jointly, but we cannot identify people in each of these states separately). To deal with this, a parent interviewed in 1978 in Michigan is matched to someone at $t = -1$ in Michigan since in 1977 we know everyone’s exact state of residence, but at all other event times, they are matched to someone within the Michigan or Wisconsin region since before the 1977 interview, state of residence is reported as Michigan–Wisconsin instead. A parent interviewed in 1976, however, is matched to someone reported to be in the Michigan–Wisconsin grouping at all event times since for all the years we are getting childless matches from, we cannot distinguish whether someone is living in Michigan or Wisconsin. On the other hand, a parent interviewed in 1982 is matched to people in Michigan at all negative event times since all the way back to 5 years before, we do know whether people are living in Michigan rather than only knowing that they live in either Michigan or Wisconsin. We really just want to maximize the observations that are matched fully on state of residence, so rather than making a clear distinction that parents interviewed before 1983 are always matched to childless people in the same state group as them and parents interviewed after this point are always matched to childless people in the same state of residence, we take into account the interview year of the childless people as well, and allow for the same parent to get matches in some negative event times based on state groups and other event times based on state of residence depending on the year their matches are interviewed in.
- Since the ACS data begins much later than the CPS data in 2000, to preserve as much data as possible, rather than dropping parents between 2000–2004 as described in [the sample selection](#), we instead match parents in ACS in these years to childless people in CPS data from 1995–1999. Therefore, parents in the first five years of available data are really only dropped in the CPS matching.

Matching identifiers (CPS and ACS)

`match_id` For parents, this variable is the original survey id; for non-parents, this is the original survey id of the parent they were matched to. *Note:* each matched non-parent appears in the data n times, where n is the number of matches. Each of these copies gets the original survey id from a different matched parent.

matched0–matched5 For non-parents and parents with children aged 1–10, these variables are always NA. For parents with children aged 0, **matched i** is 1 (0) if this parent is matched to some non-parents i years younger and interviewed i years before.

dataset & matching_sample The variables **dataset** $\in \{\text{CPS, ACS}\}$ and **matching_sample** $\in \{\text{Weekly, Annual}\}$ together identify from which matching sample each observation comes from.

match_doiy For parents, this variable is equal to the own date of interview. For non-parents, this variable is equal to the date of interview of the parent they were matched to. *Note:* similar to the note above.

match_doby Same as **match_doiy**, only for date of birth.

Final Pseudo-Panel Dataset

To be able to run an event study using the cross-sectional data, we have to create some variables in the CPS and ACS data after running the matching. The general principle is that since we are creating a pseudo-panel, all the childless people in the pre-period are in effect considered to be the same as their match and we would like to use the information provided by their match on everything except labor market outcomes and number of children (Since childless people are matched to someone at event time 0 they will always be matched to someone with 1 child only, so we cannot use the same principle here). By construction, they will be the same on the variables we use to match (education level, race, marital status, state, gender) but for all other variables we need to take values from the matched observation with a child for every childless person. This will be the case for variables like age at first birth, metro status, state of birth etc.

Variables to keep original observed values for

1. Earnings
2. Weekly Employment
3. Annual Employment
4. Number of children
5. Age
6. Indicator for any children
7. Age of oldest child in household
8. Survey weights

New variables to be created which use matched value for childless people and observed value for parents

1. Demographic variables: Female, all education variables, marital status, all race variables, year of birth of respondent
2. Fertility: Age at first birth, year of birth of first child, decade of birth of first child
3. Geography: State of residence, state groups, census division of residence, election results in state of residence, metro status, country and state of birth, census division of birth
4. Interview details: five-year interval of interview, decade of interview
5. Indicator for the dataset data comes from

Note: some of these variables are used in the matching procedure so the observed value for childless people and the value for the parent they are matched to will be identical.

Variables to create matched pseudo-panel version for and also keep original version

1. Identifier for individual, interview year

Other variables to create

1. Event times: we need to create a separate event time variable for Weekly Employment and Earnings/Annual Employment since Weekly Employment refers to the week before interview while Earnings and Annual Employment refer to the previous year.
 - (a) Weekly employment: This is equal to age at the time of interview – age at first birth for all individuals who are matched as part of the ACS and CPS Weekly Employment matching samples and missing for the observations in the separate Earnings matching (since most or all of the Earnings matched sample will have Weekly Employment observations and will be repeated between both samples and we do not want to double count them in our event study analysis).
 - (b) Earnings/Annual Employment: This is equal to age at the time of interview – age at first birth – 1 to take into account that it refers to the previous year. It is treated as missing for true parents with a child aged 0 at the time of interview since we only want to use our matched sample in the pre-period. We also only use the annual matched sample here and not the weekly employment matched sample. Similarly, we also create a post variable that corresponds to each event time variable that takes on a value of 0 for event times –5 to –1 and a value of 1 for event times 0 to 10.

Note: The Weekly Employment and Earnings samples can also be kept completely separate, but I found it easier to stack them in one dataset with separate event time variables that are non-missing only for the particular matched sample they pertain to.

2. Matching weight: For childless people, this is equal to $1/(\text{the number of childless people matched to their matched parent})$ and for parents this is just equal to 1 (we want a weight to account for the number of available matches for each parent so each childless person does not count as 1 whole observation if there were ties in that bin, but we want all parents to be included as 1 whole observation)
3. Regression weights: We then need to create new regression weights that combine the survey weights and the “matching weights” which will be the original survey weight times the “matching weight”. For example, a parent with a survey weight of x will get a regression weight of $x * 1$, while one of the n childless people they were matched to at a particular event time who has a survey weight of y will get a regression weight of $y * \frac{1}{n}$.

2.2 Matching (PSID and NLSY)

In order to validate the matching procedure used on CPS and ACS, we run the matching on the PSID and NLSY panels as well, using the same basic method described above for CPS and ACS with key adjustments for the fact that these datasets are actual panels with *observed* negative event times.

1. **Run four matchings.** PSID (Annual), PSID (Weekly), NLSY (Annual), NLSY (Weekly). Annual matchings drop individuals with missing earnings; Weekly matchings drop individuals with missing weekly employment.
2. **Annual vs. Weekly construction of negative event times.** For each of the Annual and Weekly matchings, take individuals at event time $t = 0$ *based on the corresponding event-time variable* (earnings/annual-employment event time for the Annual matching; weekly-employment event time for the Weekly matching). For each parent at $t = 0$, match to childless individuals who are 1 year younger and interviewed 1 year earlier to obtain $t = -1$, 2 years younger and 2 years earlier to obtain $t = -2$, and so on up to $t = -5$.
 - **Biennial interviewing adjustment.** In later years, interviews are conducted every other year in both PSID and NLSY. If interview year $y - 1$ is unavailable, we use the nearest available prior interview year while preserving the correct age lag. For example, for someone interviewed in 2019 in the PSID, we cannot match to anyone in 2018 for $t = -1$, but we can use 2017: for $t = -1$, match to someone 1 year younger in 2017; for $t = -2$, match to someone 2 years younger in 2017; for

$t = -3$, match to someone 3 years younger in 2015; etc. In general, when $y - k$ does not exist in the panel, replace it by the closest available $y - k'$ *strictly less than* y , and adjust the younger-age offset accordingly so that the implied event time remains $-k$.

- **No $t = 0$ matching in panels.** For CPS/ACS we anchor on “child aged 0” and then create $t = -1$ using a same-year, same-age childless match (for Annual) because earnings refer to the prior calendar year. In PSID/NLSY, event time is defined using exact dates of birth and interview months, and the annual series already accounts for the lag in earnings. Therefore we do *not* match $t = 0$ to a same-age, same-year childless observation in the panels.
3. **Childless donor pool.** The pool consists of: (i) individuals never observed with a child anywhere in the panel (event time always missing), and (ii) individuals at *any* negative event time, including $t < -5$. For the purpose of constructing pseudo negative event times, we ignore whether panel information indicates eventual fertility when $t < 0$, so that we can compare pseudo-assigned t to the true panel t .
 4. **Variables to match on.** Same as CPS/ACS: gender, education, race/ethnicity (with Hispanic as a separate category), marital status, and state of residence. **Exception:** in NLSY, state of residence is not available for matching and is therefore excluded.
 5. **Weights.** As in CPS/ACS, create matching (tie) weights equal to $1/n$ when a parent has n eligible childless matches at a given negative event time. Form the regression weight as the product of the survey weight and the matching weight.
 6. **Event-study construction.** The pseudo-event studies based on matched PSID and NLSY data are constructed exactly as the actual PSID/NLSY event studies, and child penalties are estimated in the same way as described above. Parents at $t = 0$ are kept even if no childless match is found for them, as in CPS/ACS.
 7. **Sample restrictions.** Use the same age-at-first-birth restriction as in CPS/ACS and drop parents interviewed in the first five survey years, so that $t = -5, \dots, -1$ can be constructed.

Matching identifiers (PSID and NLSY)

The following variables are created for the PSID/NLSY matching:

match_id For parents, this variable is the original survey ID. For non-parents, this is the original survey ID of the parent they were matched to. *Note:* each matched non-parent appears in the data n times, where n is the number of distinct parent matches; each copy inherits the **match_id** of a different matched parent.

matched-5 – matched-1 Same logic as in CPS/ACS, with two exceptions: (i) the PSID/NLSY naming convention uses hyphens (e.g., **matched-5**), which corresponds to **matched5** in CPS/ACS; (ii) there is *no* **matched0** in PSID/NLSY because, unlike CPS/ACS, we do not match people at $t = 0$ to same-age, same-year childless observations.

dataset and matching_sample As in CPS/ACS, **dataset** $\in \{\text{PSID}, \text{NLSY79}\}$ is provided. The variable **matching_sample** used in CPS/ACS is *not* created here. However, Weekly vs. Annual samples can be identified because within the Weekly (Annual) sample the weekly (annual) event-time variable is non-missing while the annual (weekly) event-time variable is missing.

match_doiy For parents, equal to their own date (year) of interview. For non-parents, equal to the interview date (year) of the matched parent. Repeated non-parent rows inherit the corresponding parent's value for each match.

match_doby Same as **match_doiy**, but for date (year) of birth.

match_ch1yob For parents, equal to the date (year) of birth of the first child. For non-parents, equal to the date (year) of birth of the first child of the parent they were matched to. *Note:* each matched non-parent appears n times, each copy inheriting the first child's birth year from a different matched parent.

3. Results

This section contains details on the empirical specifications used to produce the results in “Results for Replication”.

3.1 General Notes for Results

Regressions and weighting

Notes for all regressions: For everything before the epidemiological approach in the paper, use the “regression weights” that we have defined above in all regressions as probability weights (called “wgt” in my dataset), which is equivalent to using them as analytic weights with robust standard errors. Note that for results using an actual panel rather than a pseudo-panel, since we don’t have any matched observations and additional weights to consider, the regression weights are equal to the survey weights and the matching weights can simply be treated as 1. For the epidemiological studies in the paper, we instead use the “matching weights” (called “wgt_match” in my dataset) for regressions instead because we are more constrained in terms of power.

Even if not specified below, note that whenever we are taking averages or performing any kind of analysis using childless people we have to take the matching weights into account (i.e. each childless person can NEVER be treated as a single observation unless they were in fact the only available match for their parent) but we might not always be using the survey weights as well. The counterfactuals (which will be described later) are always based on an average that uses the matching weights, but not the survey weights. All regressions are run using feols in the fixest package in R.

Please note that for all analyses using the pseudo-panel where we estimate child penalties across various dimensions (such as across time, space etc.), for observations in the pre-period, rather than using people who report that they live in a specific state or were interviewed in a specific time interval, we use childless people who were matched to the parents in the group we are interested in. For example, if we are estimating child penalties in the time interval 2015-20, the sample will be composed of parents interviewed in 2015-20 and childless people who were matched to those parents from 2015-20. This is why I re-define certain variables after the matching and use the value of the parent childless people were matched to instead of keeping their original value, so when using my data, it is possible to run an event study for parents in 2015-20 in the way described above simply

by subsetting the 5-year interval variable rather than having to subset parents based on their year of interview and then also having to identify their childless matches.

Binscatters

This contains some general information regarding the binscatters in Figures 11, 14, and 17. We always split the x-axis variable into deciles and then plot the average of the y variable within each decile against the average of the x variable within each decile. The red shaded areas represent confidence intervals on the prediction around each dot from an OLS regression bootstrapped 200 times using the function "boot" from the package boot. The function essentially runs an OLS regression on 200 different subsamples of size 10 from the sample data (with replacement) and then you need to predict the y value for each of the original 10 dots on the graph (not the 10 dots boot is using for the OLS regression). Boot will then provide confidence intervals on the predictions which is what we plot on the graphs. The slope and standard error included on the graph are from a simple linear regression using the 10 plotted dots rather than from a bootstrapped regression. Before running the bootstrapping set a seed of 9746290.

Also note that whenever data is split into groups such as deciles/quintiles/quartiles etc such as in all the binscatter graphs, R functions are used to automatically conduct the splitting which will sometimes be affected by the ordering of the data, especially in cases where we have many ties. In general, R automatically determines the split in cases when there are ties based on the way the data is ordered, which in my data will be by factor level (implies alphabetical order for character variables and chronological order for any time variables), but if there are specific differences between results in cases like this, the decisions made regarding ties should be the first place to consider. The default behavior of these functions in R in cases where decile/group size must be uneven is to distribute more observations to the lower deciles, and for our purposes we continue following this default option. Any other specific decisions made regarding decile splits will be specified in the sections they pertain to. Use unrounded numbers as much as possible to prevent dealing with ties.

Heatmaps

The legends for all heatmaps are constructed by splitting the data into deciles rather than by choosing a manual scale. The rules for splitting the data into deciles discussed in the description of binscatter graphs also applies here. I use the R package "usmap" to create all heatmaps.

3.2 Descriptive Statistics Tables

This section is used to create Tables 1-3.

3.2.1 Table 1

This table uses pooled CPS and ACS data for people aged 20-50 prior to matching (using the data before we add any sample restrictions such as dropping people with missing or zero weights). Take unweighted averages of all the specified variables, separately for men and women with and without children. Redefine cohort here so that rather than using the reported year of birth variable you use the difference between each individual's interview year and age. Note that the race variable used includes hispanic as a separate category (the fraction black and white reported are all non-hispanic).

3.2.2 Table 2

This table is similar to Table 1 but we now use the matched data instead and we compare men and women at $t = 0$ to those at $t = -1$. Take averages weighted by the matching weights (but not survey weights) of all the specified variable, redefine cohort here again so that rather than using the reported year of birth variable you use the difference between each individual's interview year and age. Note that the race variable used includes hispanic as a separate category (the fraction black and white reported are all non-hispanic). Another important note is that since we have different samples for annual vs. weekly variables, the annual and weekly employment rates and the earnings averages are based on the estimation sample for each individual variable. The averages for all other variables and the number of observations are all based on the weekly employment sample since this is where we have the most power. To illustrate the matching procedure, we also drop any parents at $t = 0$ who we are unable to find matches for at $t = -1$ only for the purposes of this table. The averages are weighted by the matching weights (weights that account for the number of matches for each parent) but NOT by the regression weights which also take the survey weights into account.

3.2.3 Table 3

This table focuses on the samples of people we use in the epidemiological approach sections so it will be better to construct this table after running that analysis. Movers are defined as US-born individuals living in a different state than where they were born, while immigrants are foreign-born individuals living in the US. Each group is divided by the child penalty in their place of birth (top vs bottom quartile of child penalties in US states and foreign countries, respectively). The child penalties used to split movers by state of birth are annual employment

penalties in the sample of stayers (as presented in Figure A.15), while the child penalties used to split immigrants by country of birth are those used in Figure A.17 and throughout [this section](#). All the estimates in panel A use the annual employment sample of movers while all the estimates in panel B use the pooled employment sample of immigrants. The row for fraction living in high-penalty states is the fraction in each sample living in states in the top quartile of the distribution of states using the same annual employment penalties in the sample of stayers from Figure A.15. The fractions of the sample that are Black and White reported here are based on different variables compared to Table 1 and 2 (we include any Black or White Hispanic people in these numbers).

3.3 Validation of Pseudo-Event Study Approach

3.3.1 Figure 1

This section is used to create Figure 1 and Appendix Figures A.1-A.3 in the paper.

We always consider three main labor market outcomes- earnings, annual employment, and weekly employment. For each of these outcomes, denoting the outcome of interest by Y_{ist} for individual i at event time t , we run the following regression separately for men and women on pooled CPS and ACS data.

$$Y_{it} = \alpha' \cdot D_{it}^{Event} + \beta' \cdot D_{it}^{Age} + \gamma' \cdot D_{it}^{Year} + v_{it} \quad (3.1)$$

where we include a full set of event time dummies, omitting the event time dummy at $t=-2$, and X_i is a vector of controls (dummies for the age of the parent and dummies for calendar year)

We convert the estimated effects into percentages by calculating $P_t^g \equiv \frac{\hat{\alpha}_t^g}{E[\tilde{Y}_{it}^g]}$, where $\tilde{Y}_{it}^g$ is the average predicted outcome from the event study regression omitting the contribution of the event dummies (the “counterfactual”) and plot these estimates and 95% confidence intervals for men and women in our event study graphs. (Note that the counterfactual is obtained from the same regression and you do not have to run a different one without event times. You need to use the same event-study regression but get the predicted outcome based ONLY on the fixed effects and not the event times). The counterfactuals are weighted by the matching weights (“wgt_match”) but not by the regression weights. We then define the child penalty as $E[P_t^{men} - P_t^{women} | t \geq 0] - E[P_t^{men} - P_t^{women} | t < 0]$. **This is the average effect across treated event times net of the average effect across untreated event times. In general anywhere we have an estimate labeled “child penalty”, the treated event times over which the mean is calculated is**

the full post-period (event times 0-10). In graphs where the long-run child penalty is specified instead, the treated event times over which the mean is calculated are event times 5-10 and they are always calculated this way unless stated otherwise. From this point forward, “Child Penalty” refers to the estimate across the full period and “Long-Run Child Penalty” refers to the long-run penalty described above.

We validate our matching procedure by running the same specification on pooled panel data (PSID and NLSY) and comparing the child penalty estimates from pooled CPS and ACS data to the panel data estimates. **For any panels based on PSID and NLSY data, the child penalty is calculated as $E[P_t^{men} - P_t^{women}|t \geq 0]$, where the average effect across untreated event times is NOT subtracted as it is with CPS and ACS.**

For the appendix, we also run the following alternative matching specifications on pooled CPS and ACS data where we conduct the matching as explained above but begin with a baseline specification that matches on fewer variables and add one demographic variable at a time to view the improvement in the event study graphs as the matching is done on a more and more granular level until we reach our chosen specification. Please note that panels A-C are much more demanding than panel D and require too much server capacity to run so these specifications are currently run with a seed where each parent is randomly matched to only one childless person at each event time rather than matching with ties as we do in panel D.

1. Gender, age, year
2. Gender, age, year, education
3. Gender, age, year, education, marital status
4. Gender, age, year, education, marital status, race
5. **Chosen specification:** Gender, age, year, education, marital status, race, state

3.3.2 Figure 2

The right panels of Figure 2 are identical to those in Figure 1 (actual event studies using PSID and NLSY data). The left panels are created in the same way as the right, but using the PSID and NLSY pseudo-panel. Note that the child penalty will be calculated in the same way for the PSID and NLSY pseudo event study as it is for the actual PSID and NLSY event study (without a pre-period adjustment).

3.3.3 Hausman Test Numbers

To calculate the Hausman statistic, we need three components: the transpose of the difference in regression coefficients $(b_{\text{diff}})'$, the inverse of the difference in variance-covariance matrices $(vcov_{\text{diff}})^{-1}$, and the difference in regression coefficients (b_{diff}) . (The first component is a 1 x k matrix, the second is a k x k matrix, and the third is a k x 1 matrix, so we end up with a 1 x 1 scalar for our Hausman statistic.)

We calculate the difference in the estimated coefficients between the actual and pseudo models as:

$$b_{\text{diff}} = b_{\text{actual}} - b_{\text{pseudo}}$$

where b_{actual} is a 1 x k matrix containing the coefficients from the actual model, and b_{pseudo} is a 1 x k matrix containing the coefficients from the pseudo model.

Next, we calculate the difference in the variance-covariance matrices between the actual and pseudo models:

$$vcov_{\text{diff}} = vcov_{\text{actual}} - vcov_{\text{pseudo}}.$$

We use the `vcov` function in R to extract the variance-covariance matrix from a regression output. Then, we compute the inverse of this matrix using the `solve` function in R :

$$(vcov_{\text{diff}})^{-1} = \text{solve}(vcov_{\text{diff}})$$

We calculate b_{diff} similarly as stated above, except where b_{actual} and b_{pseudo} are k x 1 matrices.

Finally, we multiply b_{diff} with $(vcov_{\text{diff}})^{-1}$, and multiply this result with b_{diff} . (We use matrix multiplication, symbolized by `%*%` in R.) Then, we take the absolute value of the final result to ensure a positive statistic¹

The regression we run to produce the estimated coefficients we use above is similar to equation (1) in the US Pseudo paper, except where we use a post dummy, and we interact all the terms with a dummy for being a woman (aka `gender`):

$$Y_{it} = \alpha \cdot Post_{it} + \beta \cdot D_{it}^{Age} + \gamma \cdot D_{it}^{Year} + (\alpha_W \cdot Post_{it} + \beta_W \cdot D_{it}^{Age} + \gamma_W \cdot D_{it}^{Year}) * Woman,$$

where α_W is the coefficient of interest.

¹The source code for the `phtest` function in R, which runs the Hausman test, takes the absolute value to ensure the Hausman statistic is always a positive number. [See here](#)

We run this regression separately using the actual panel data (PSID/NLSY actual) and the pseudo panel data (PSID/NLSY pseudo)² to extract the actual and pseudo coefficients. We also run it separately for each outcome variable, so we end up with three Hausman statistics (one for each outcome variable). Please note that:

- Since the coefficient of interest is a scalar, the three components of each Hausman statistic will all be scalars.
- When extracting the variance-covariance matrix from a regression, be sure to only keep the value of interest (the one that involves post interacted with gender).

Keeping this in mind, we calculate the Hausman statistic as described above. Additionally, we calculate the p-value for each Hausman statistic by comparing it to the chi-squared distribution using the following code in R:

```
pchisq(value, df = 1, lower.tail = FALSE),
```

where `value` is the value of the calculated Hausman statistic, `df` is the degrees of freedom (1 in this case), and `lower.tail` is set to `FALSE` as is done in the source code for the `phtest` function in R.

3.3.4 Table A.1

Table A.1 empirically tests Assumption 3 (conditional independence) by comparing outcomes between the actual panel and pseudo-panel data. We use both the actual panel, constructed by combining PSID and NLSY79 data, and the pseudo-panel derived from pooled PSID/NLSY observations.

First, we identify and handle non-matched parents to ensure comparability between the two panels. Non-matched parents—those with no matches at any $\tau < 0$ or matched at only one $\tau > 0$ —are dropped from the actual panel at all event times and from the pseudo-panel only at $\tau \geq 0$. This creates four analysis samples combining panel type and time event variables.

To facilitate interpretation and comparison across different subgroups, we scale earnings by the mean counterfactual within the actual panel, separately by sex. The mean counterfactual is the weighted mean of fixed effects from regression (3.1) on the actual panel. We calculate point estimates by running regression (3.1) for each *panel* $\times$ *outcome* $\times$ *sex* combination. For each $\tau < 0$, we compute

²You can also use the CPS/ACS pseudo data instead, but this produced worse results.

$$\hat{\beta}_\tau^g + \mathbb{E} \left[\tilde{Y}_{it}^g \mid \tau_{it} = \tau \right]. \quad (3.2)$$

Table A.1 presents two versions of this statistic for each *panel* $\times$ *outcome* $\times$ *sex* combination: the value at $\tau = -2$ (where $\hat{\beta}_\tau^g = 0$ as the omitted base year) and the simple average across all $\tau < 0$.

Beyond individual estimates, we calculate three types of differences. First, for each *outcome* $\times$ *sex* combination, we compute the difference between panels (pseudo minus actual). Second, for each *panel* $\times$ *outcome* combination, we compute the sex difference (men minus women). Third, for each outcome, we calculate the difference-in-differences: how the sex gap differs between the pseudo and actual panels.

We obtain standard errors through bootstrap replication with 200 iterations, drawing rows with replacement from each dataset. We use seed 72839251 to ensure replicability.

3.3.5 Figure 3

Figure 3 validates the pseudo-event study approach in subsamples using both the actual panel, constructed by combining PSID and NLSY79 data, and the pseudo-panel derived from pooled PSID/NLSY observations. We run the following regression separately for each *panel* $\times$ *stratification-variable* $\times$ *outcome* $\times$ *sex* combination:

$$Y_{it} = \alpha' \cdot D_{it}^{SV} + \beta' \cdot (D_{it}^{SV} \times D_{it}^{Post}) + \gamma' \cdot (D_{it}^{SV} \times D_{it}^{Age}) + \phi' \cdot (D_{it}^{SV} \times D_{it}^{Year}) + v_{it}, \quad (3.3)$$

where the stratification variables are geography (4 census regions), time (5 decades), education (high school or less vs college), marital status (single vs married), and race (4 categories). The specific census regions, decades, and race categories are reported in Figure 3 in the paper. D_{it}^{SV} is a vector of dummy variables, each equal to 1 iff observation it belongs to the respective category of the stratification variable SV . D_{it}^{Post} is a dummy equal to 1 iff $\tau_{it} \geq 0$. The coefficient vector of interest is β .

For each category of SV , we scale the corresponding element of β by the mean counterfactual among all observations it belonging to that category with $D_{it}^{Post} = 1$. We then compute the child penalty for each SV category as the difference in these scaled β coefficients between men and women (men minus women).

Finally, for each outcome, we plot the child penalties across all categories of all stratification variables estimated from the pseudo panel against those from the actual panel. We also calculate the R^2 from regressing the pseudo panel child penalties on the actual panel child penalties, including an intercept.

3.3.6 Figure A.4

Once the pseudo-panel using PSID and NLSY is complete, we create Appendix Figure A.4 which is a histogram that compares event times obtained from the matching to the actual event times observed in the PSID and NLSY. We calculate the difference in event times (predicted - actual) for everyone that the pseudo-event study has predicted to be between event times -5 to -1. For people who never have children in the panel, this difference is essentially equal to positive infinity. They are included in the histogram bin labeled "Never". However, for the histogram only, we exclude people in the matched sample who are only observed in the panel before the age of 45 since they may still have children (this is to exclude young people who we don't have an event time for because of attrition rather than because they never actually had children or people who enter the sample in later years and are too young to have children yet in the panel interview period). The "Never" bin also includes anyone with an event time difference greater than 20. We use the annual matching results only to make the histogram. The histogram is also weighted by the matching weights so that childless people are not double counted.

3.3.7 Figure A.5

The right panels of Figure A.5 plot the weighted average of estimates across four age-at-first-birth groups: 25–28, 29–32, 33–36, and 37–45. These results are obtained by estimating the main regression equation in the paper, adding the age group as dummies. The estimates shown in the panels represent the weighted average of the group-specific estimates, where the weights correspond to the proportion of the sample in each age-at-first-birth group. Specifically, weights are calculated as frequency weights—the number of observations it in each age-at-first-birth group within the full sample, divided by the total number of observations it in the full sample. Observations with missing values for the outcome, event time, age, age-at-first-birth, or year are excluded from these counts.

The standard error for the weighted average is calculated using the following formula:

$$SE = \sqrt{w_1^2 \times SE_1^2 + w_2^2 \times SE_2^2 + w_3^2 \times SE_3^2 + w_4^2 \times SE_4^2}$$

where w_i denotes the sample weight for group i , and SE_i is the standard error of the estimate for group i .

3.3.8 Child Penalties over Time

This section is used to create Figure 4 and Appendix Figures A.6-A.10 in the paper.

We run the following regression separately for men and women to estimate child penalties within 5-year intervals

$$Y_{ity} = \eta' \cdot D_{ity}^{Interval} + \lambda' \cdot D_{ity}^{Event \cdot Interval} + \beta' \cdot D_{ity}^{Age \cdot Interval} + \gamma' \cdot D_{ity}^{Year \cdot Interval} + v_{ity} \quad (3.4)$$

where the specification follows from Equation (3.1), but we interact the full set of event dummies with a dummy for each time interval. The intervals are 1973-79, 1980-84, 1985-89, 1990-94, 1995-99, 2000-04, 2005-09, 2010-14, 2015-20.

As explained [here](#), note that each time-interval dummy takes on a value of 1 if the individual is a parent interviewed within the interval or a childless individual matched to a parent interviewed within the interval and takes on a value of 0 otherwise. The pre-period sample always includes those matched to true parents, and this principle applies to all analyses split by time period. In my pseudo-panel dataset, the variable called “yr5” is used to define these time intervals.

We calculate percentage impacts for gender g in time interval y as

$$P_{ty}^g \equiv \frac{\hat{\lambda}_{ty}}{E[\tilde{Y}_{ity}^g | t]} \quad (3.5)$$

Since event time dummies are interacted with 5-year interval dummies, the above definition of the percentage impact in each time interval only holds if there is no omitted category. If you choose to omit a 5-year interval, you will need to add multiple regression coefficients to get the full effect in all the five-year intervals except the omitted one. This will add an extra step to estimate the standard error in each interval, so I recommend running the regression as specified above without an omitted time interval, or running separate regressions for each time interval instead of including interactions at all.

Additional note for Figure A.9: For the unscaled earnings penalty and counterfactual, nominal earnings growth rates are calculated as follows: compute the growth rate of average nominal earnings in CPS data among the working-age population (25–60) from the mid-year of the respective time interval (e.g., 2017 for the 2015–2020 interval) to the year 2020. The average is weighted by the regression weights (`wgt`).

3.4 Child Penalties over Space

3.4.1 Event Studies and Heatmaps of Penalties

This section is used to create Figure 5-7 and Appendix Figures A.11-A.14.

We run the following regression separately for men and women to estimate child penalties by state:

$$Y_{its} = \eta' \cdot D_{its}^{State} + \lambda' \cdot D_{its}^{Event \cdot State} + \beta' \cdot D_{its}^{Age \cdot Censusdivision} + \gamma' \cdot D_{its}^{Year \cdot Censusdivision} + v_{its} \quad (3.6)$$

where the specification follows from Equation (3.1), but we interact the full set of event dummies with a full set of dummies for each of the 50 US States and the District of Columbia in order to estimate separate event time coefficients for each state. Since we are more constrained when we run this specification within state, rather than interacting age and calendar year dummies by state as well (equivalent to running a separate regression for each state), we interact the fixed effects with census division dummies. Some states are reported in groups before 1977, so we simply drop those FIPS codes in our state analysis, but keep data before 1977 for states that are always reported on their own. (In practice, what we do is run separate regressions as specified by (3.6) for each census division, but alter the equation slightly to leave age and calendar year dummies uninteracted, mainly because R does not provide the correct results if fixed effects are interacted by a different category than the main predictors, but depending on the package used to run regressions, it may be possible to run (3.6) directly. What this means is that you will need to run 9 regressions to get these results (1 per census division), and each regression will include interactions for all the states in that census division, but will not include any states outside that census division.).

We calculate percentage impacts differently as follows $P_{ts}^g \equiv \frac{\hat{\lambda}_{ts}}{E[\hat{Y}_{its}^g | t]}$.

Same as for the results by time interval, you do not want an omitted state dummy in these regressions.

The child penalties are estimated as described [previously](#), and the same penalties are split into deciles and then used to create heatmaps in Figure 5, with darker colors representing higher penalties.

3.4.2 Empirical Bayes Correction

This section is used to create Appendix Figures A.15-A.16, which compare OLS and Empirical Bayes (EB) shrinkage estimates of state-level child penalties. Appendix section B of “Gulek_WP_2024” is a great place to start for understanding the use of Empirical Bayes Correction.

We start by first running separate event study regressions across states using fixed effects for age-gender and year-gender cells. Outcome is regressed on state interacted with gender, state, post interacted with state, the interaction of post, gender, and state, with fixed effects for the interaction of age and gender, interaction of year of interview and gender, age, year of interview. State-level estimates are then obtained, and Empirical Bayes shrinkage is applied by estimating the variance of the true signal relative to the variance of the estimation error. Each state's estimate is shrunk toward the national mean proportionally to its noisiness, with more noisy estimates undergoing greater shrinkage.

Two sets of plots are produced for each outcome: the first set compares OLS and EB shrinkage estimates side-by-side with confidence intervals, and the second plots EB shrinkage estimates against OLS estimates along a 45-degree line.

3.4.3 Raw Gender Gaps and Counterfactual Employment Rates

3.4.4 Fraction of Gender Gaps Explained by Penalties

This section creates Appendix Figure A.10

We estimate the raw gender gap in each outcome as $Gap_y \equiv \frac{E[Y_y^{men}|t \geq 0] - E[Y_y^{women}|t \geq 0]}{E[Y_y^{men}|t \geq 0]}$ where $E[Y_y^g|t \geq 0]$ is the weighted average outcome for parents of gender g in time period y (the weights are the same survey weights used in all regressions). The fraction explained is then defined as $\frac{CP_y * (E[\tilde{Y}_{ity}^{women}|t \geq 0] / E[Y_y^{men}|t \geq 0])}{Gap_y}$, where CP_y is the child penalty in time period y , $E[Y_y^{men}|t \geq 0]$ and Gap_y are as described above, and $E[\tilde{Y}_{ity}^{women}|t \geq 0]$ is the unweighted average predicted outcome for mothers (event times 0-10) from the event study regression in Equation (3.4) omitting the contribution of the event dummies. We then plot a time series of the fraction explained in each 5-year interval for each outcome.

3.4.5 Scatterplots of Child Penalties and Counterfactuals

This is used to create Figure 7.

One concern with our child penalty estimates is that since we adjust the results from the regression by the counterfactual, the variation we see could be driven by the counterfactuals themselves. In order to address this, Figure 7 goes back to the regressions in equation (3.6) and estimates unadjusted child penalties in each state. Everything will be done in the exact same way, except the counterfactual adjustment will not be made at all. Figure 7 then plots this unadjusted child penalty against the adjusted child penalties we have been working with otherwise in the left panels. In the right panels, it plots the counterfactual employment rates for mothers

against the adjusted child penalties. The counterfactual rates are defined as the unweighted average predicted outcome from the event study regressions, omitting the contribution of event times for mothers.

3.5 Child Penalties by Demographics

This section is used to create Figure 8 in the paper.

In each section of the figure, we run separate regressions for men and two samples of women. Men are never split based on demographics since childbirth is a non-event for them while women are split based on demographics.

In all of these graphs, we report the long-run penalty for the separate samples of women as defined [here](#), where for both groups the penalty is calculated relative to the series of men shown in the graph. 95% confidence intervals are shown.

3.5.1 Education

Low-educated women are defined as those with a high school degree or less and high-educated women have a college degree or more. Equation (3.1) is run separately for these subsamples of women and for men with either a high school degree or less or a college degree or more together (note that this means that both men and women with “some college” are not included at all).

3.5.2 Marital Status

Equation (3.1) is run separately for single women (never married, separated, divorced, or widowed), married women (all others) and for married and single men together using the married dummy.

3.5.3 Race

Equation (3.1) is estimated separately for white women, black women, and for men (white and black combined). Individuals classified as “other race” in the CPS/ACS race variable are excluded entirely from the analysis. We use the CPS/ACS *race* variable, which has three categories (white, black, other), rather than the *race/ethnicity* variable that includes an additional Hispanic category.

3.6 Child Penalties by Fertility (for slides only)

Use Equation (3.1) to run regressions for women with 0-1 children, 0-2 children, and any number of children (this corresponds to our baseline specification in the full CPS/ACS sample), and all men separately. Plot all 4 series on one graph and report the long-run penalty for women in each group measured relative to all men.

3.7 Child Penalties vs GSS Data

We create a GSS dataset with averages at the state level, decade level, and state-decade level as described in the cleaning, but to deal with missing state-decade combinations in the collapsed GSS state-decade dataset we develop a method to impute these missing values.

3.7.1 Imputing GSS Values

Since many state and decade combinations are missing, we use the GSS state-decade dataset previously created and create an alternate version that fills in these missing values.

1. Use the GSS state by decade dataset
2. Drop any missing GPI values before calculating percentiles in the following step
3. Within each decade, calculate the percentile from that each GPI value corresponds to
4. Calculate the average percentile by state
5. Impute each missing state-decade GPI observation with the average percentile calculated for that state within that decade.
6. For example: If GPI is missing for Alabama in 1972-79 and available in all other decades, for each decade we estimate the percentile that Alabama is at and take the average, calling this value x . We then calculate what GPI value is represented by the x^{th} percentile using the distribution in 1972-79 and assign Alabama in 1972-79 that GPI value.

This gives us a final dataset with a GPI observation for every state-decade combination. A time series for each state is plotted in Figure A.14, where imputed observations are indicated by empty dots.

Note that all results using the GPI are based on this new dataset with imputations. We collapse this decade at the state level and at the decade level separately to get the following results.

3.8 Heatmap of Gender Norms

Figure 9 is created by taking unweighted averages of the GPI by state using GSS state-decade data.

3.8.1 Child Penalties vs GSS Over Time

Generate a dataset which takes a weighted average of the GPI by states and decade in each decade, where the weights for each state are the fraction of the US population living in that state, calculated using the 2019 Census Bureau Population Data (popweights_2019.rds). These numbers are plotted against the child penalties over decades which are estimated like the penalties [here](#) but by decades instead of by five-year interval. Then plot the GPI by decade against the child penalty by decade in Figure 10. Decades can be identified using variable "yr10" in my CPS/ACS pseudo-panel where the 70s range from 1973-1979, 80s are 1980-89, 90s are 1990-99, 00s are 2000-09, 10s are 2010-20.

3.9 Child Penalties by State and Time

Since we are much more constrained in terms of power when we estimate penalties by state as well as time interval, we do not run a full event study as in other specifications. Instead, we run the following regression for men and women separately and for each decade separately (1973-79, 1980-89, 1990-99, 2000-09, 2010-20):

$$Y_{its} = \alpha' \cdot D_{its}^{Post} + \eta' \cdot D_{its}^{State} + \lambda' \cdot D_{its}^{Post \cdot State} + \beta' \cdot D_{its}^{Age} + \gamma' \cdot D_{its}^{Year} + v_{its} \quad (3.7)$$

where *Post* takes on a value of 1 if an individual has a child ($t \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$) and 0 if the person has been matched to a parent ($t \in \{-5, -4, -3, -2, -1\}$) and age and calendar year fixed effects only vary by gender and decade but not by state.

We then define the percentage impact in the post period as $P_t^{gy} \equiv \frac{\hat{\alpha}_t^{gy} + \lambda_s^{gy}}{E[Y_{it}^{gy}|t]}$. Since we are not including event time dummies here, the child penalty in each decade is simply defined as $P_t^{my} - P_t^{wy}$.

3.10 Child Penalties vs GSS Data Across States and Time

This section creates Figure 11 in the paper.

We run the following specifications, all of which use the [previously estimated child penalties over space and time](#). Child penalties are topcoded at 1 for this analysis. All plots include a line of best fit for the binned data

that is ultimately plotted and text with the slope from the main regression specification.

State Fixed Effects: Regress child penalties in state s and decade d on GPI values as follows

$$CP_{s,d} = \alpha_{s,d}GPI_{s,d} + \gamma_s + v_{s,d} \quad (3.8)$$

where γ_s represents state fixed effects. We then residualize the child penalty values by the state fixed effects and add the average state fixed effect coefficient across all states back to each observation so that the adjusted child penalty value is $RCP_{s,d} \equiv CP_{s,d} - \hat{\gamma}_s + \frac{\sum_{i=1}^{51} \hat{\gamma}_i}{51}$. We then split the GPI values into deciles and calculate the mean adjusted child penalty within each GPI decile and plot them against the average GPI within that decile. The slope reported on the panel is the slope of the line of best fit on the binscatter rather than the slope from this regression.

State Fixed Effects and Demographic Controls: Regress child penalties in state s and decade d on GPI values as described above but add controls and then residualize by all the fixed effects and controls in the specification. The controls are estimated at the state and decade level within the estimation sample for each variable separately and are all weighted. The controls are the fraction of women with a high school degree or less, the fraction of women with a college degree or more, the fraction of women who are married, the fraction of white women in the sample, and the fraction of black women in the sample (where hispanic people are included within their racial category). As above, after child penalty estimates are residualized, we add the average state fixed effect back to the child penalties. We also do the same for all the other controls by adding back the average effect of each control in the sample which is the average of the control variable across all states and decades multiplied by the coefficient in the regression. As above, the mean adjusted child penalty is calculated within each GPI decile and is plotted against the average GPI in each GPI decile. The controls used can be found in my dataset called “gss_controls_gpi_vs_penalties.rds.”

3.11 Epidemiological Approach: US-Born Movers

For the following analysis, the sample is restricted to individuals whose state of birth we know, which is only in the ACS. Movers are defined as those whose state of birth is different from their state of residence, while stayers are defined as those with the same state of birth and residence. We focus only on annual and weekly employment, and we use the “matching weight” to weight all regressions going forward.

We run 3 event studies: separately for all US-born men, stayer women, and mover women, including all 51 states together as in Equation (3.6), but do not interact age and calendar year fixed effects with anything (they

are estimated across the US rather than within state or census division). Another key difference is of course that we use state of birth rather than state of residence. The event studies are plotted in Appendix Figures A.18-A.19 where we include the series for all men in a given state and two separate series for stayer and mover women in the given state. The child penalties for mover and stayer women listed on the panels are relative to the series for all US-born men shown in the graph. Figure 12 presents a subset of these state event studies while Figure 13 plots the child penalties for movers and stayers in each state against each other.

The right panels of Figure 13 control for selection on state of residence using the residualization method described in Section 6.2 of the paper. To implement this, we first calculate the fraction of movers from state s residing in state s' , denoted $\alpha_{ss'}^r$, as the ratio of the weighted number of mothers who moved from s to s' to the weighted total number of mothers who moved from s . These fractions use matching weights and include only observations with non-missing values for outcome, event time, age, and year. The predicted child penalty for movers from state s is then computed as $\widehat{CP}_s^{movers} = \sum_{s'} \alpha_{ss'}^r CP_{s'}^{stayers}$. We run the regression $CP_s^{movers} = \beta \cdot CP_s^{stayers} + \gamma \cdot \widehat{CP}_s^{movers} + \nu_s$ and residualize the mover penalties by calculating the sum of the intercept and $\beta \cdot CP_s^{stayers}$. The slopes, standard errors, and adjusted R^2 annotated in the right panels come from this regression, while the regression lines shown match those in the corresponding left panels.

Figure A.20 follows an analogous procedure but splits movers by state of residence rather than state of birth. Here, the fractions represent movers residing in state s who moved from state s' , calculated as the ratio of the weighted number of mothers who moved from s' to s to the weighted total number of mothers who are movers and reside in s . The same weighting criteria and residualization steps apply.

Child penalties for stayers for the pooled employment outcome (described in the following section) should also be estimated, as we will use those child penalties in the epidemiological approach for foreign immigrants.

3.12 Epidemiological Approach: Foreign Immigrants

For the following analysis, the sample is generally restricted to foreign-born individuals whose country of birth we know, which is in the ACS for all years and the CPS from 1994 onwards. We also focus on a subset of countries of interest. Since we are constrained in terms of power, we also focus on a new outcome, namely pooled employment. The pooled employment outcome simply combines annual and weekly employment observations, so for a given person, if we have both annual and weekly employment available, we treat each one as a separate observation in the event study. Therefore, all observations with a non-missing annual event time or a non-missing weekly event time will be used. We also use the “matching weight” to weight all regressions in this analysis.

For each foreign country, we run Equation (3.1) separately for men and women who report the country as their country of birth to create Figure 14 and Appendix Figure A.21. Each panel in these figures reports the child penalty calculated as usual based on these event studies (labeled and hereafter referred to as “US Immigrant Penalty”), as well as the child penalty estimated within these countries using data from a different project (labeled and hereafter referred to as “Country of Birth Penalty”). I will give you a dataset with these penalties from the different project - you will not need to estimate them. Please note that the child penalties in each country of birth in that dataset are always bounded at 0 (so any negative penalties should be rounded up to 0). Also note that we treat Scandinavia as one country for this analysis, where we pool the observations of Denmark, Norway, and Sweden.

To create Figure 15, the child penalties in the country of birth for all chosen countries are split into deciles. Indicators for the decile of penalty in the country of birth is added to our CPS/ACS dataset and we run event studies for US immigrants for each decile separately. The “Country of Birth” penalty reported is as weighted average of the penalties of the countries in those deciles where the weights for each country are its share of US immigrants in the CPS/ACS data in each decile.

Panel A of Figure A.22 simply plots the child penalties in each country of birth and penalties for US immigrants from each country as described above (the penalties on each panel in Appendix Figure A.21) with a line of best fit. Panel B plots the same penalties; however the line of best fit is based on a regression with controls for education, marriage, race, fertility, age at first birth, and US location across immigrant mothers from different countries. The controls are the fraction of people with a high school degree or less, fraction with a college degree or more, fraction married, fraction white, fraction black (where hispanic people are included in their particular racial category), average number of children, average age at first birth among parents, the fraction of people living in low penalty states, the fraction of people living in high penalty states. All the controls are calculated using the sample of mothers born in each country in the pooled employment sample and are weighted using the matching weights. The fraction of people in low and high penalty states is defined as the fraction of people living in states in the bottom and top quartile of the distribution of child penalties for stayers, respectively. The child penalties for stayers used are those calculated [here](#) for **pooled employment**. As we do [in the Penalties vs GSS Binscatters](#), we regress the child penalties for immigrants against the child penalties in the country of birth with the set of controls and then residualize the immigrant penalties by the estimated effect of the controls for each country. The average effect of controls across all countries is then added back to the residualized outcome.

Figure 16 is Figure A.21 in binscatter form - we create deciles of child penalty in birth country and plot this against the average child penalty for immigrants within that decile. The slope shown is the slope of the line of

best fit shown on the graph.

Appendix Figure A.23 is created the same way as the education part of Figure 8, but for immigrants only.

The sample used to create Figure A.24 is slightly different from the sample used for the analysis in the rest of this section: rather than focusing only on foreign-born individuals, we also use US-born people of foreign descent. The sample of immigrants used here is defined as foreign-born individuals from the 81 countries of interest ("first-generation") as well as individuals with ancestry from the 81 countries of interest who are born in the US ("later-generation"). Men are not split based on which generation of immigrant they are, but we do restrict to either first or later-generation immigrant men only. Women are then split by their generation of immigration and the figure is then constructed in the same way as Figure 8. Panel A and B are created by splitting the child penalties in country of birth into quartiles and then running the event studies separately for people born in/with ancestry from countries in the bottom quartile and top quartile separately.

Please note that in this section, any time we split the data into deciles of birth-country penalty, since we have 81 countries, and therefore will not have an even number of countries in each decile, the first decile will have 9 countries and the remaining deciles will have 8 countries. R should automatically do this, but basically you want to make sure that all the deciles containing more countries are the ones with smaller birth-country penalties. There will be countries with tied birth-country penalties as well, so to decide on ties, re-order the dataframe by our estimated child penalty for immigrants before assigning to deciles (i.e. if South Africa and the Philippines have an equal child penalty in their country of birth, but need to be in separate deciles, the country with the smaller penalty for immigrants in the US is assigned to the lower decile).